

LUCIDPREP

Grade 12 Mathematics

U.S. Standards-Aligned Course Syllabus

PRECALCULUS

Precalculus · Common Core (+)

36 weeks • 4 modules • 70% mastery

Curriculum Architecture

This course is organized as 36 weeks across four 9-week modules. Every week follows LucidPrep's established secondary learning cycle: AI Instructor opening, objectives, vocabulary, five warm-ups, main lecture, two short tutorial videos, AI Instructor presentation, two homework sets, and a 40-question assessment.

Mastery standard: 70%+. Students below mastery review the instructional presentation and may retake the assessment, matching the existing LucidPrep secondary model.

U.S. Alignment Note

A fourth year of mathematics is recommended for college readiness but is not required in every state. Precalculus is the most common Grade 12 course for college-bound students; others take Calculus, Statistics, or a quantitative-reasoning course. This syllabus uses the Common Core high-school (+) standards as the primary reference.

Prepared for LucidPrep • Curriculum Planning Edition • 2026

Common Instructional & Assessment Model

Weekly Learning Cycle

Step	Component	Description
1	Opening / Domain Overview	~2-minute AI Instructor introduction
2	Objectives + Vocabulary	Student-facing outcomes and core terminology
3	Warm-Up	5 questions with explanations
4	Main Lecture	Concept instruction, models and examples
5	Video Tutorials	2 short instructional videos
6	AI Instructor Presentation	Visual models, examples, strategies and applications
7	Homework	Two sets of 5 questions
8	Weekly Assessment	40 questions; 70% mastery
9	Review / Retake	Below mastery → review presentation → retake

Assessment Philosophy

Assessment is continuous and mastery-based. Each week pairs low-stakes practice (five warm-ups and two homework sets) with a 40-question weekly assessment. Students scoring below 70% review the instructional presentation and retake, so mastery is reached before progressing. Module assessments at Weeks 9, 18 and 27 and a cumulative final at Week 36 measure retention and synthesis.

PRECALCULUS

Grade 12 mathematics aligned to the Common Core State Standards (+ standards)

This 36-week course is designed around **the Common Core high-school (+) standards: Functions (F-IF, F-BF, F-TF), Trigonometry, Complex Numbers, Vectors & Matrices (N-CN, N-VM) and conic sections — preparing students for calculus**. It is a full-year curriculum with four nine-week modules and a consistent LucidPrep mastery workflow.

Grade	Grade 12
Course	Precalculus · Common Core (+)
Duration	36 weeks
Structure	4 modules × 9 weeks
Weekly assessment	40 questions
Mastery threshold	70%
Module assessments	Weeks 9, 18, 27
Final assessment	Week 36

Course Learning Outcomes

- Analyze and transform functions and their inverses across families.
- Extend trigonometry to identities, equations and the unit circle.
- Operate with complex numbers, vectors and matrices.
- Model with conic sections, sequences and rates of change.
- Build the conceptual foundations for calculus.

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Module 1 — Functions & Their Behavior

Week	Topics	Students will be able to	Standards
1 — Function Families & Transformations	Parent functions; transformations; symmetry	Analyze and transform functions	F-BF.3
2 — Combining & Composing Functions	Operations; composition; domain	Combine and compose functions	F-BF.1c
3 — Inverse Functions	Existence; finding; restricting domain	Find and verify inverses	F-BF.4
4 — Polynomial & Rational Functions	Analysis; asymptotes; end behavior	Analyze polynomial and rational functions	F-IF.7
5 — Exponential & Logarithmic Functions	Modeling; equations; base e	Analyze and apply exp/log functions	F-LE.4
6 — Piecewise & Advanced Graphs	Piecewise functions; continuity (informal)	Analyze piecewise functions and continuity	F-IF.7b
7 — Modeling with Functions	Choosing and fitting models to data	Model data with appropriate functions	F-BF.1
8 — Rates of Change	Average rate of change; secant lines	Analyze rates of change	F-IF.6
9 — Module 1 Assessment	Cumulative review	Demonstrate mastery of Module 1	F-IF/F-BF

Module delivery notes. Emphasize functions as the central object and prepare for calculus through rates of change, informal limits and rigorous work with trigonometric and transcendental functions.

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Module 2 — Trigonometry

Week	Topics	Students will be able to	Standards
1 — Angles & Radian Measure	Unit circle; radians; arc length	Work with radian measure	F-TF.1
2 — Trig Functions & the Unit Circle	Definitions; special angles	Evaluate trigonometric functions	F-TF.2
3 — Graphs of Trig Functions	Amplitude, period, phase/vertical shift	Graph and transform trig functions	F-TF.5
4 — Inverse Trig Functions	Definitions; domains; solving	Use inverse trig functions	F-TF.6/7
5 — Trigonometric Identities	Fundamental, Pythagorean, even/odd	Prove and apply identities	F-TF.8
6 — Sum, Difference & Double-Angle	Deriving and applying formulas	Apply advanced identities	F-TF.9
7 — Trigonometric Equations	Solving over intervals	Solve trigonometric equations	F-TF.7
8 — Laws of Sines & Cosines	Oblique triangles; modeling	Solve triangles and model periodicity	G-SRT.10/11
9 — Module 2 Assessment	Cumulative review	Demonstrate mastery of Module 2	F-TF

Module delivery notes. Emphasize functions as the central object and prepare for calculus through rates of change, informal limits and rigorous work with trigonometric and transcendental functions.

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Module 3 — Complex Numbers, Vectors & Matrices

Week	Topics	Students will be able to	Standards
1 — Complex Numbers & Operations	Arithmetic; conjugates; modulus	Operate with complex numbers	N-CN.3
2 — Polar Form & De Moivre's Theorem	Polar representation; powers and roots	Use the polar form of complex numbers	N-CN.4/5
3 — Vectors	Magnitude; direction; components	Represent and interpret vectors	N-VM.1/2
4 — Vector Operations	Addition; scalar multiplication; applications	Operate with vectors	N-VM.4/5
5 — Matrices & Operations	Add, subtract, multiply; scalar	Operate with matrices	N-VM.7/8/9
6 — Matrices & Transformations	Representing transformations; determinants	Use matrices for transformations	N-VM.11/12
7 — Solving Systems with Matrices	Inverse matrices; systems	Solve systems using matrices	A-REI.9
8 — Applications	Vectors and matrices in modeling	Apply vectors and matrices to problems	N-VM
9 — Module 3 Assessment	Cumulative review	Demonstrate mastery of Module 3	N-CN/N-VM

Module delivery notes. Emphasize functions as the central object and prepare for calculus through rates of change, informal limits and rigorous work with trigonometric and transcendental functions.

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Module 4 — Analytic Geometry & Foundations for Calculus

Week	Topics	Students will be able to	Standards
1 — Conic Sections — Parabolas	Focus/directrix; equations	Analyze parabolas	G-GPE.2
2 — Ellipses & Hyperbolas	Equations; foci; graphs	Analyze ellipses and hyperbolas	G-GPE.3
3 — Parametric Equations	Representing curves and motion	Use parametric equations	F-IF
4 — Polar Coordinates	Polar graphs; converting	Graph in polar coordinates	N-CN
5 — Sequences & Series	Arithmetic/geometric; summation; convergence	Analyze sequences and series	A-SSE.4
6 — Limits (Informal)	Approaching values; one-sided; continuity	Reason about limits informally	F-IF.6
7 — Introduction to the Derivative	Instantaneous rate; tangent lines	Interpret the derivative as a rate	F-IF.6
8 — Area & Accumulation (Preview)	Area under a curve; accumulation	Reason about accumulation informally	modeling
9 — Final Assessment	Cumulative final across all modules	Demonstrate mastery of the full course	(+) standards

Module delivery notes. Emphasize functions as the central object and prepare for calculus through rates of change, informal limits and rigorous work with trigonometric and transcendental functions.

Assessment & Alignment

Assessment Blueprint

Warm-ups: weekly • Homework: weekly • Weekly assessment: 40 questions • Module assessments: Weeks 9 / 18 / 27 • Final: Week 36 • Below 70%: instructional review and retake.

Standards Alignment

Primary framework	CCSS (+) standards: F-IF, F-BF, F-TF, N-CN, N-VM; conic sections
Mathematical Practices	Make sense of problems and persevere; reason abstractly and quantitatively; construct viable arguments and critique reasoning; model with mathematics; use tools strategically; attend to precision; use structure and regularity.
Recurring Concepts	Multiple representations; equivalence; proportional reasoning; structure; modeling; real-world application.
State mapping	Common Core adopted by the large majority of states; Texas TEKS and other state frameworks cover the same grade-level content under different codes.

Content Production Guidance

Author each week as 5 warm-up items, a concept lecture with worked examples and multiple representations, 2 curated videos, an AI presentation with models and strategies, 2 homework sets (5 items each), and a 40-question bank spanning fluency, application and reasoning. Tag every item to its Common Core code.

Curriculum Basis & Source Notes

This Grade 12 mathematics syllabus was generated from the U.S. Common Core State Standards for Mathematics, including the (+) standards. Content reflects Precalculus, the common college-preparatory Grade 12 course.

Primary reference	Common Core State Standards for Mathematics — high-school (+) standards in Functions, Trigonometry, Number & Quantity (complex numbers, vectors, matrices) and analytic geometry, and the Standards for Mathematical Practice.
State cross-check	California Common Core and Precalculus course standards.
State cross-check	Texas Essential Knowledge and Skills (TEKS) for Precalculus.
LucidPrep alignment	Mapped to LucidPrep's 36-week, 4-module structure with 70% mastery.

Recommended next production step. Commission item banks per week (40 assessment items plus warm-ups and homework), each tagged to its Common Core code, then load into the LucidPrep authoring pipeline.